Exact or Nothing

Gate-Governed Measurement of the Liquidation Map
on a Transparent Perpetual-Futures Venue

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Abstract

Research on liquidation cascades is usually conducted on a liquidation map that has been *estimated*—from aggregate open interest, an assumed leverage distribution, and a reconstructed population of positions. We ask what becomes possible, and what becomes visible, when the map is instead *observed*: on Hyperliquid, per-position liquidation prices are published, the mark price is built from an oracle the venue itself publishes, and any address's fill history is public. We describe an instrument built on two rules—*exact or nothing* (a quantity that cannot be had exactly is not computed, because the approximation is what makes a backtest lie) and *gates before models* (every command downstream of a validation gate refuses to run when that gate is missing, stale, or failing, and the refusal is the product). The instrument's substrate is a write-once capture store in which every record carries both an event timestamp and an ingest timestamp, making lookahead a queryable property rather than a code-review hope, and a hash-chained ledger in which a failed verdict cannot quietly leave the record. We then report what the instrument measured. Its feed gate passes twenty checks; its map gate does not pass, and we explain why that is the informative outcome. Registry coverage is 24.7–38.0% of venue open interest by symbol under a conservative side convention; the derived liquidation price reproduces the venue's published value to within 10^{-4} for 40.8% of 2,926 positions, with an unexplained residual we decline to model away. The central finding is negative and, we believe, general to any venue where positions must be sampled: of 48 observed liquidations, the notional-seeded registry contained 2.9% of the liquidated *wallets* but 21.9% of the liquidated *notional.* Mass and events are carried by different populations, so a sampling scheme optimized for one is mis-specified for the other, and no shape-preserving rescaling repairs the difference. We report that the mapped fraction is currently 0.0%, show that this figure is confounded by snapshot ordering rather than informative, and specify the scheduled experiment that discriminates among the three explanations. We present no performance figures; the contribution is an instrument, a discipline of refusal, and four measurements that change what the observational obstacle is understood to be.

Keywords: market microstructure, liquidation cascade, perpetual futures, decentralized exchange, measurement, research infrastructure, causality, reproducibility, coverage bias, sampling

JEL Classification: C55, C81, G14, G17

Contents

1 Introduction

A leveraged perpetual-futures position carries a deterministic liquidation price. In aggregate these prices form a spatial field—a *liquidation map*—and the structural claim, developed formally in the companion paper [10], is that when price reaches a dense region of this field, forced market liquidations impose impact that can carry price to the next dense region, producing a self-reinforcing cascade. The claim is intuitive, mechanically plausible, and hard to test, for a reason that is observational rather than theoretical: on most venues the map cannot be seen. Position-level data is private; what is published is aggregate open interest. The map is therefore reconstructed by assuming a leverage distribution over that aggregate, and every downstream result inherits the assumption.

This paper reports on an instrument built to remove the assumption rather than refine it. Hyperliquid is, to our knowledge, the venue where the fewest such assumptions are necessary: it publishes a liquidation price per position, it liquidates against a mark price built from an oracle it also publishes, funding is settled hourly and observable, its backstop vault's inventory is visible, and the fill history of any address can be read by anyone. On such a venue, the interesting question is not how to estimate the map better. It is what the map—and the research programme built on it—looks like when estimation is refused outright.

1.1 Two rules

The instrument is organized around two rules, and both are enforced in code rather than intended in prose.

Gate	Question	What a FAIL kills
`feed`	is the captured data intact and causal?	everything
`map`	does the map predict the venue's own liquidations?	the map branch
`magnet`	does the cluster pull, net of costs, beat a baseline?	the magnet experts
`persistence`	does skill in $[t-90d,\, t]$ predict $(t,\, t+30d]$?	the smart-money branch
`decay`	does cohort-flow alpha survive past minutes?	copyability

Table 1: The gate ladder. Each gate is permitted to terminate the branch beneath it; two of them are permitted to terminate the project. Ordering them before the modelling work is deliberate.

Exact or nothing. Where a quantity cannot be obtained exactly, it is not computed. A position that is not observed is absent from the map and enters the reported coverage deficit; it is not imputed. There are no estimated maps, no assumed leverage mixes, and no aggregator history. The justification is asymmetric: an approximation that is 80% right yields a backtest that is confidently wrong, whereas an acknowledged hole yields a coverage number, and a coverage number can gate a decision.

Gates before models. The research programme is expressed as a small set of experiments that are each *allowed to end the project.* Each is a command that writes a PASS or FAIL verdict to an append-only ledger, and every command downstream of a gate refuses to run while that gate is missing, stale, or failed. The refusal is not a safety feature bolted onto a research tool; it is the primary output. A system that will happily produce a Sharpe ratio from data whose integrity check failed has not produced a Sharpe ratio.

1.2 Contributions

1. A capture and validation substrate in which *lookahead is queryable*: every record carries the venue's event clock and our ingest clock, and a replay audit fails on any feature that read a record it could not have known (Section 3).

2. A refusal protocol built on a hash-chained ledger, with the property that a failed verdict cannot be quietly re-run away, and a formal statement of the condition under which a downstream command is permitted to execute (Section 4).

3. A liquidation map whose coverage is a first-class, printed, gate-thresholded quantity rather than an unstated assumption, together with the side-convention question that makes it a factor of two (Section 5).

4. Four measurements, of which the important one is negative: the population that holds notional and the population that gets liquidated are different, and the gap between them is an order of magnitude (Sections 6–7).

We report no trading performance. The instrument's map gate does not currently pass, and under the second rule that fact bars the downstream work rather than being reported alongside it.

2 Why a Single Transparent Venue

Restricting to one venue is usually a limitation. Here it is load-bearing, and the reason is that the alternative silently reintroduces the estimation the first rule exists to forbid.

A cross-venue design would supply global price pressure from a large centralized exchange and liquidation geometry from the decentralized one. That design requires a second clock, a second cost model, and an alignment assumption between them. The single-venue design does not, because the venue publishes the oracle price against which it constructs its own mark price. The premium

$$\pi_t \;=\; \frac{p_t^{\text{mark}} - p_t^{\text{oracle}}}{p_t^{\text{oracle}}} \tag{1}$$

is then a native, exactly observable measure of exactly the global pressure a second-venue sidecar would have been introduced to supply—and the causal arrow points the right way, since global price drives liquidations against a map we observe directly rather than infer.

Three further quantities have no clean analogue elsewhere and materially shape what is measurable. First, funding is settled *hourly*, which at cascade horizons is a first-order carry term rather than a rounding error. Second, the venue operates a backstop liquidity vault whose inventory change, Δ_t^{HLP}, is public; when forced flow overwhelms the resting book the vault absorbs it, so Δ_t^{HLP} *measures* the portion of a cascade the book failed to absorb. Absorption exhaustion, which elsewhere must be inferred from price behaviour, is here directly observed. Third, a single address's perpetual, spot, and vault activity are all visible together, so an intra-venue hedge is detectable rather than an unmodelled confound in any flow-following study.

What is given up, and we state it rather than minimize it: liquidation density on other venues (only ever an estimate, which the first rule rejects in any case); a long history spanning many regimes, which is handled with wider embargoes and smaller models and never by borrowing history from another venue; the thin-alt cascade tail; and the operational risk of depending on one venue, which is accepted explicitly.

3 Capture: Making Lookahead Queryable

Every result downstream depends on the claim that a feature computed for time T used only information available at T. That claim is normally defended by code review. We make it a property of the data.

3.1 Two clocks

Each captured record carries two timestamps: t^{event}, the venue's clock, and t^{ingest}, the first moment at which we could have known the record. The causality contract is then a filter rather than a convention.

Definition 3.1 (Causal readable set)**.** For a bar closing at time T, the admissible input set is

$$\mathcal{F}_T \;=\; \big\{\, r \;:\; t^{\text{ingest}}(r) \le T \,\big\},$$

and a feature is *causal* if it is measurable with respect to $\mathcal{F}_T$.

Definition 3.1 is enforced by replay: the audit recomputes features under the filter and fails on any difference from the production value. The distinction from $t^{\text{event}} \le T$ matters precisely when it is most dangerous—a record the venue stamped before T but which reached us after T is exactly the kind of input that inflates a backtest while looking defensible.

3.2 A write-once store

Raw records are appended to per-stream files that rotate on the UTC hour. When a file closes it is hashed and an entry is written to a manifest; the file is never reopened. A restart inside the same hour begins a *new part* rather than appending to the closed one, which gives the store a useful property: a changed digest always means corruption or tampering and never merely a bounced daemon. A per-stream sequence number continues across restarts, resumed from the manifest, so a hole in the sequence means records were *lost* rather than that the process died.

The feed gate audits this structure directly. Its checks include manifest integrity (every closed file still hashes to its recorded digest), orphan detection (an unmanifested file older than two hours indicates a writer that died without closing, so its tail may be truncated), per-stream sequence continuity, ingest monotonicity, gap detection at $60\times$ the stream's cadence hint, staleness at $20\times$, and a staleness check on the rate-limit constant table itself—each limit is recorded with the date a human last verified it against the venue's documentation, and the gate warns once that stamp exceeds ninety days. A limit that moved without anyone noticing is exactly the failure that stamp exists to surface.

3.3 Reconstruct, do not poll

The registry cannot be maintained by polling. Measured on 2026-08-07, a full sweep of clearing-house state across 3,596 wallets takes approximately five minutes and consumes the entire per-IP request-weight budget of 1,200 units per minute, leaving nothing for the trade tape. Polling frequency would therefore *become* the map's time resolution—an infrastructural constraint silently determining a scientific one.

Positions are instead maintained between snapshots by reconstruction from the public fill stream, with polling demoted to reconciliation on a rotating sample, and drift beyond tolerance failing the feed gate. Cross-margined and isolated-margined positions take separate paths, because a cross-margined account liquidates as a unit: one position's move relocates that account's entire contribution to the map. The bootstrap limitation is stated plainly—the tape supplies *changes* while snapshots supply *levels*, so a wallet is reconstructable only from its most recent snapshot forward.

4 The Gate Protocol and the Refusal Ledger

4.1 The ledger

Gate verdicts, specification freezes, run records, and periodic observations are appended to a hash-chained log. Writing $e_i = (i,\, \tau_i,\, \kappa_i,\, \pi_i)$ for the i-th entry's sequence number, timestamp, kind, and payload, the chain is

$$h_i \;=\; \text{SHA-256}\big(\text{enc}(i,\, \tau_i,\, \kappa_i,\, \pi_i,\, h_{i-1})\big), \qquad h_{-1} = 0^{64}, \tag{2}$$

with enc a canonical (key-sorted, separator-fixed) JSON encoding so that the digest is reproducible. Verification walks the chain and reports the first index at which the sequence, the predecessor link, or the recomputed digest disagrees.

The construction is standard [5,,8] and the threat model is not an adversary. It is the researcher. Equation (2) makes deletion of an inconvenient result detectable, so the fourteen variants tried before the one that worked remain on the record. This is the mechanism by which multiple-testing discipline becomes structural rather than remembered—a concern developed at length for the feature and process layers in [11, 13].

4.2 The refusal predicate

Definition 4.1 (Admissibility)**.** Let $V(g)$ denote the most recent verdict for gate g, with fields $\text{pass}(g) \in \{0, 1\}$ and timestamp $\tau(g)$. A command depending on g may execute at time t if and only if

$$V(g) \text{ exists } \wedge\; \text{pass}(g) = 1 \;\wedge\; t - \tau(g) \leq \Delta,$$

with Δ a freshness limit defaulting to 24 hours. Otherwise the command raises a refusal naming which of the three conditions failed.

The third clause is the one that is easy to omit and expensive to omit. A verdict describes the data as it stood when the gate ran; past some age it says nothing about the data about to be used. Without a freshness bound, a gate that passed once becomes a permanent licence.

Remark 4.1 (A gate that cannot be evaluated must not pass)**.** Some criteria are unanswerable on demand. The map gate's predictive criterion requires map snapshots *followed by* observed liquidations, so it is unanswerable until capture has accumulated both. The gate reports insufficient history and FAILs. Treating an unevaluable criterion as passing—or as not applicable—is the single most convenient error available in this design, and it would convert the whole protocol into decoration.

Remark 4.1 is why the results in Section 6 include a failing gate and no performance numbers. That is the protocol operating as designed, not a gap in the work.

5 The Map and Its Coverage

The map bins observed positions by liquidation price relative to the current mark, recording notional per bucket, the cross-margined share separately, and position counts; band aggregates $L^{\uparrow}(b)$ and $L^{\downarrow}(b)$ are taken over $b \in \{0.5, 1, 2, 5\}\%$. The geometry and the cascade mechanics built on it are the subject of [10]; here we are concerned with the number that qualifies the field.

Definition 5.1 (Coverage)**.** For coin c at time t, with N_t the total notional of observed positions and OI_t the venue's reported open interest,

$$\widehat{\rho}_t \;=\; \frac{N_t}{s \cdot \mathrm{OI}_t}, \qquad s \in \{1, 2\}, \tag{3}$$

where s is the side convention.

The convention deserves its own sentence because it is a factor of two on the one quantity by which the map is judged. Under the standard one-sided reading, an exchange counts each contract once in OI_t; but every long has a matching short, and a registry that tracks wallets sees both sides. Total position notional outstanding is then $2\,\mathrm{OI}_t$, and the conservative choice is $s = 2$. We take it, print it on every map alongside the coverage figure, and list it as an open verification item rather than resolving it by preference.

A map without its coverage number is, in the design's own phrasing, a lie with a chart on it. The number is not decoration: the map gate thresholds on it at $\widehat{\rho} \geq 0.25$.

5.1 Observing realized liquidations

Scoring the map requires realized liquidation events, and finding them is less obvious than it appears. The public trade tape carries no liquidation marker. The venue's fill records carry one in two places: on the liquidated wallet's own fill, in a direction string, and—far more usefully—on the *counterparty's* fill, as a structured object naming the liquidated user, the mark price, and the method (market or backstop).

The asymmetry between the two is extreme. Measured on 2026-08-08 over 117,485 fills from 70 wallets, exactly *one* fill carried the liquidated-side direction string, while 16 of the 70 wallets had taken the other side of somebody's liquidation. The efficient observer set is therefore not the registry at all: it is a handful of high-volume market makers, because whoever absorbs forced flow sees it. Since the fill endpoint has no per-user cap over the request interface—unlike the streaming interface's limit of fifteen users per connection—this scales with the weight budget rather than with connection count.

Quantity	Value	Provenance
`gate feed` verdict	PASS, 20 checks, 0 failed	ledger, 2026-08-07
`gate map` verdict	FAIL on `predictive`	ledger, 2026-08-08
Coverage, BTC	0.315	ledger, $s = 2$
Coverage, ETH	0.380	ledger, $s = 2$
Coverage, SOL	0.247	ledger, $s = 2$
Derivation exact fraction	0.408 of 2,926 positions	rel. err. $< 10^{-4}$
Derivation median rel. err.	0.152 (p90: 0.921)	vs. published value
Liquidated wallets in registry	2.9% of 35 wallets	ledger observation
Liquidated notional in registry	21.9% of $223,491	ledger observation
Mapped notional fraction	0.0% *(confounded, Section 7)*	ledger observation

Table 2: Measurements to date. The map gate fails; under Remark 4.1 this is the correct verdict rather than a missing result.

Remark 5.1 (The lookahead trap in scoring)**.** An event that predates the position snapshot it is scored against is not a prediction; it is a memory. Scoring drops such events, and reports the number dropped rather than absorbing it silently. Of the first 48 events observed, 24 predated the snapshot and were correctly excluded on these grounds—a figure worth reporting precisely because a scorer without this check would have counted them.

6 Measurements

All figures below are read from the instrument's ledger and are reported with the date on which the verdict or observation was appended. Nothing in this section is a backtest.

6.1 Coverage

A leaderboard exposes 41,392 wallets with equity and windowed volume—roughly forty times the population that trade-stream discovery reaches, which is what makes a seeded registry a different regime rather than a larger version of the same one. The two natural seeds behave differently, and the difference turns out to matter more than the levels (probe of 2026-08-07): the top 2,000 wallets *by equity* span 61% of venue open interest but only 28% hold any position at a given moment, while the top 2,000 *by weekly volume* span 39% with 46% active. Their union reaches 69%. Accordingly the map registry seeds by equity, since it wants whoever holds size, and the skill cohort seeds by volume, since it wants whoever trades.

The operational registry's realized coverage under Definition 5.1 with $s = 2$ is 0.315 for BTC, 0.380 for ETH, and 0.247 for SOL. We note without resolving it that the ratio between these figures and the 69% scoping probe is close to s itself, which is consistent with the probe having been computed one-sided; the convention remains an open verification item, and we report both numbers rather than adopting the more flattering one.

6.2 Fidelity of the derived liquidation price

The venue publishes a liquidation price per position, and that published value is the source of truth. A derived value is needed only for positions whose size changed since the last snapshot and which therefore have no current published number. Deriving is thus an approximation with a measured error, not a replacement—and measuring the error is the point.

Diffed against the published field over 2,926 positions: median relative error 0.152, ninetieth percentile 0.921, and agreement to within 10^{-4} for 40.8%. Restricted to accounts holding no isolated margin, agreement rises to 58% over 1,074 cross positions. The residual is unexplained.

Size-tiered maintenance margin on large positions is the leading suspect, and the map gate declines to fail on fidelity alone—the derivation is a maintenance approximation, not the primary source—but it does fail on a *collapse* below 20%, which would indicate that the venue's margin rules had moved while the fill-maintained map continued blind.

We draw attention to this as a small instance of the first rule paying for itself. A design that derived liquidation prices unconditionally would have shipped a map that is materially wrong for roughly three positions in five, with nothing in the system able to say so.

7 The Population Mismatch

Measuring coverage was expected to close the observational problem posed in [10]. It did not. It exposed a second problem, different in kind, and the paper's central finding is that the second one is not repairable by the standard remedy for the first.

7.1 Two distinct biases

The usual correction for partial observation assumes the observed field is a uniform thinning of the true one, so that rescaling by OI_t/N_t restores magnitude while leaving every ratio feature untouched. The assumption is that the sampling is unbiased in *shape*. That is what fails here, and it fails for a structural reason:

> The registry is seeded by equity, because the map wants whoever holds size. But the wallets that get liquidated are small and highly leveraged. Sampling proportional to notional is precisely *not* sampling proportional to liquidation events.

Of 48 liquidations observed and appended to the ledger, the registry contained 2.9% of the liquidated *wallets* (35 distinct addresses) and 21.9% of the liquidated *notional* ($223,491). A second reading one cycle later gave 2.8% and 21.9% over 49 events. The order-of-magnitude gap between the two percentages is the second bias stated quantitatively, and it is invisible to any coverage figure of the form (3), which is denominated in notional and therefore reports the flattering one of the two.

The finding cuts both ways, and we resist collapsing it to either. Cascades are driven by notional, and catching a fifth of liquidated notional is not nothing; per-position scoring is not obviously dead. What the field is poor at is seeing the long tail of small forced exits—which matters exactly to the extent that those small exits are what *initiates* a chain rather than what continues one.

Remark 7.1 (Coverage of what?)**.** The generalizable form of this result is that "insufficient coverage" is an incomplete diagnosis. Coverage is a ratio, and a field can be simultaneously adequate for the mass that sustains a cascade and inadequate for the events that start one. Any venue on which positions must be sampled rather than enumerated is exposed to this, and a single coverage scalar will not reveal it. We now report coverage both by notional and by event count for that reason.

7.2 Why the reading cannot yet be believed

Both readings show 0.0% of liquidated notional *mapped*—attributable to a position the map had marked in advance. Taken at face value this is a refutation of the instrument. It should not be taken at face value, and stating why is more useful than the number.

A liquidated wallet no longer holds the position. A registry snapshot taken *after* a liquidation therefore shows nothing that could have predicted it, and the current snapshot cadence does not reliably precede the events being scored. The measurement consequently cannot distinguish *"different populations"* from *"we looked too late"*. Reporting the zero without the confound would be the more striking result and the less honest one.

7.3 The experiment that discriminates

A scheduled snapshot-then-observe cycle—sweep the registry, then scan for liquidations, appending an overlap reading to the ledger each pass—converts a single confounded reading into a series, and the series separates the hypotheses. Because sweeps are expensive (Section 3), the cycle persists last-run times so that a restart does not re-trigger a sweep and a job never overlaps itself; the cadence is roughly six-hourly snapshots against hourly scans.

1. **Mapped notional fraction climbs off zero.** Per-position scoring works; the map gate's predictive criterion becomes evaluable as specified, and the instrument stands.
2. **Mapped fraction pinned at zero, notional fraction holding near** 22%**.** The registry sees the right wallets but never *before* they are liquidated. Validation must move from per-position to cluster-level scoring—did liquidations land in the price buckets the map marked, regardless of whose positions they were. A weaker claim, but answerable with the registry as it stands.
3. **Notional fraction collapses as well.** Neither; the registry needs a third seed drawn by *liquidation risk*—margin ratio, or distance to liquidation price, both computable from data already held—which targets the population that actually fires, at the cost of another sweep.

Only the first outcome validates the instrument as specified. None of the three is obtainable from further analysis: the discriminating variable is calendar time under capture, which is why capture runs while the rest is built and why day one of capture is the project's real start date. A point-in-time series cannot be recovered later.

8 Relation to the Companion Papers

This paper is the measurement layer beneath a set of related results, and the division is worth stating explicitly.

[10] supplies the *model*: the density field, the eight scale-invariant geometric features, the closed-form domino condition, and the absorption damper. It named the sample-to-population problem as its binding obstacle and proposed an open-interest rescaling. Sections 5–7 of the present paper are the first measurements against that obstacle, and they revise it: the sampling rate is now known, and the rescaling is now known to be insufficient, for a reason the earlier paper did not anticipate.

[11] establishes, on a different venue and feature set, that a strong and universal predictor can be economically inert once one conditions on the states in which a passive fill actually occurs—an adverse-selection barrier that bounds monetization independently of signal quality. That result is why the present instrument treats execution as a first-class layer with its own simulator and cost hash rather than as a post-processing step.

[13] develops the analytical contract—the *process* as a unit that certifies whether a feature carries information about a specified target—and the orthogonality mathematics for counting a signal book's effective degrees of freedom. The gate ladder of Table 1 is the same discipline applied one level lower, to the data rather than to the signals.

[12] documents the predictive structure of the order book itself, including the finding that conditioning on directionally-consistent maker fills collapses the information coefficient while conditioning on the presence of any fill raises it. Together with [11] it establishes the execution boundary this instrument's allocation and execution layers are designed around.

9 Limitations and Open Verification Items

We list what is unverified with the same prominence as what is measured, because the design treats an unverified constant as a live defect rather than a footnote.

Load-bearing constants not yet confirmed. The side convention s in (3)—a factor of two on the headline number. The liquidation formula's residual: the derivation reproduces the published value for 40.8% of positions, with size-tiered maintenance margin the prime suspect for the rest. The precise construction of the mark price and which price triggers liquidation. The funding formula, cadence, and clamp. The rate-limit weight table, which carries a verification stamp that goes stale at ninety days. The streaming connection limit per IP, which bounds any future per-user subscription work. Node-data format, retention, and access cost, which sizes the historical-backfill milestone and therefore the persistence gate.

Known measurement gaps. The order-book stream's cadence is sparser than its documented hint implies—45 records over 70 seconds across three coins—so any depth-dependent feature must wait on a proper cadence measurement. Clock skew flipped sign between capture runs ($-196\,\text{ms}$ then $+395\,\text{ms}$); both are within the two-second tolerance, but a $\sim 600\,\text{ms}$ shift over minutes is unexplained, and the candidates (NTP discipline versus the venue's timestamp semantics) have different implications for Definition 3.1.

Structural risks. Reconstruction may drift from truth between reconciliations; drift beyond tolerance fails the feed gate, which converts the risk into a detection problem rather than removing it. Cascade episodes overlap heavily, so any downstream model is exposed to the usual overlapping-label pathologies and requires purging, embargoing, and uniqueness weighting [7], with a mandatory naive baseline per model. Crowding is sharper here than elsewhere for the reason that motivates the whole design: an exactly observable map is exactly observable to hunters too. Finally, the venue's short history is handled with wider embargoes and smaller models, never by padding with another venue's data—the first rule again.

10 Conclusion

We have described an instrument for studying liquidation cascades on a venue where the liquidation map is observed rather than estimated, and reported what it measured. Two design commitments do the work. *Exact or nothing* converts unobserved positions into a reported coverage deficit instead of an imputation, which is what makes the deficit measurable at all. *Gates before models* converts validation into a refusal that binds every downstream command, which is what makes a failing verdict survive contact with the desire for a result.

The measurements are these. The feed gate passes twenty integrity and causality checks. Registry coverage is 24.7–38.0% of venue open interest by symbol under the conservative side convention. The derived liquidation price reproduces the venue's published value to within 10^{-4} for 40.8% of 2,926 positions, with an unexplained residual we report rather than model away. And the map gate fails, because its predictive criterion cannot yet be evaluated and an unevaluable criterion is not permitted to pass.

The finding we regard as general is the population mismatch. A registry seeded to capture notional captured 21.9% of liquidated notional but only 2.9% of liquidated wallets, and the gap between those numbers is invisible to a coverage statistic denominated in notional. Sampling optimized for mass is mis-specified for events, and no shape-preserving rescaling repairs the difference, because the sampling is biased in shape rather than merely thinned. We further report that the mapped fraction is currently zero, and that this figure is confounded by snapshot ordering rather than informative—a distinction that a scheduled snapshot-then-observe series will resolve in one of three specified directions, only one of which leaves the instrument's claims intact. We state the discriminating experiment in advance, in the ledger, so that whichever way it resolves, the record will show that it was specified before it was run.

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